

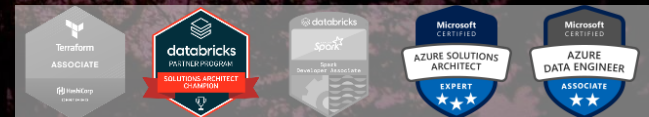


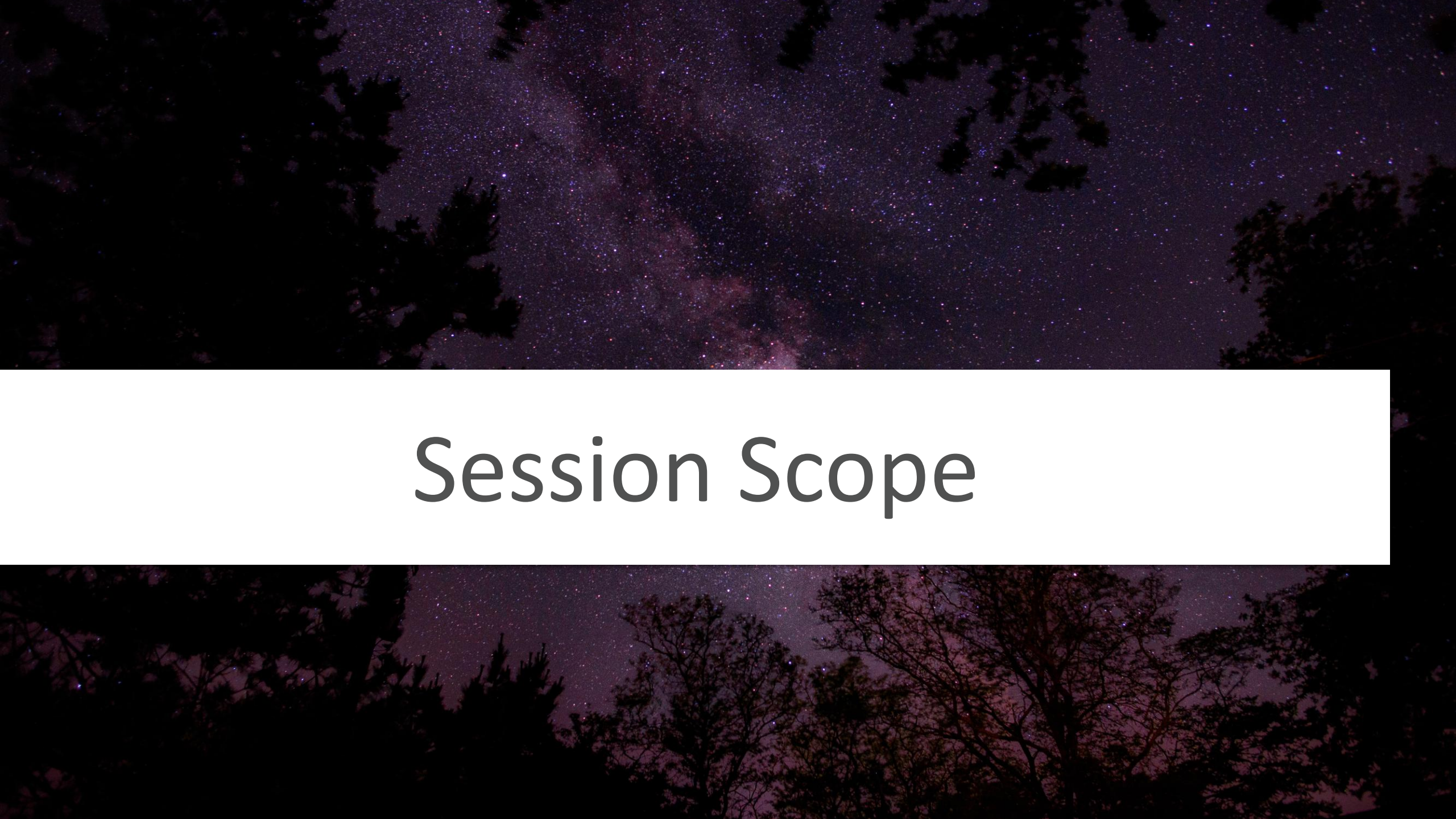
 | @falekmiah

 | falekmiah.com

 | FalekMiah01

Value of DevOps Release Process in Data Teams





Session Scope

What is Release Management?

What's the benefit of DevOps?

DevOps is for IT teams?

Why do we need Development Environments?

How can we implement CI/CD?

What tools should we use?

How do we decide on Branching Strategy?

...



Stakeholders



Lack of Understanding

- Various backgrounds



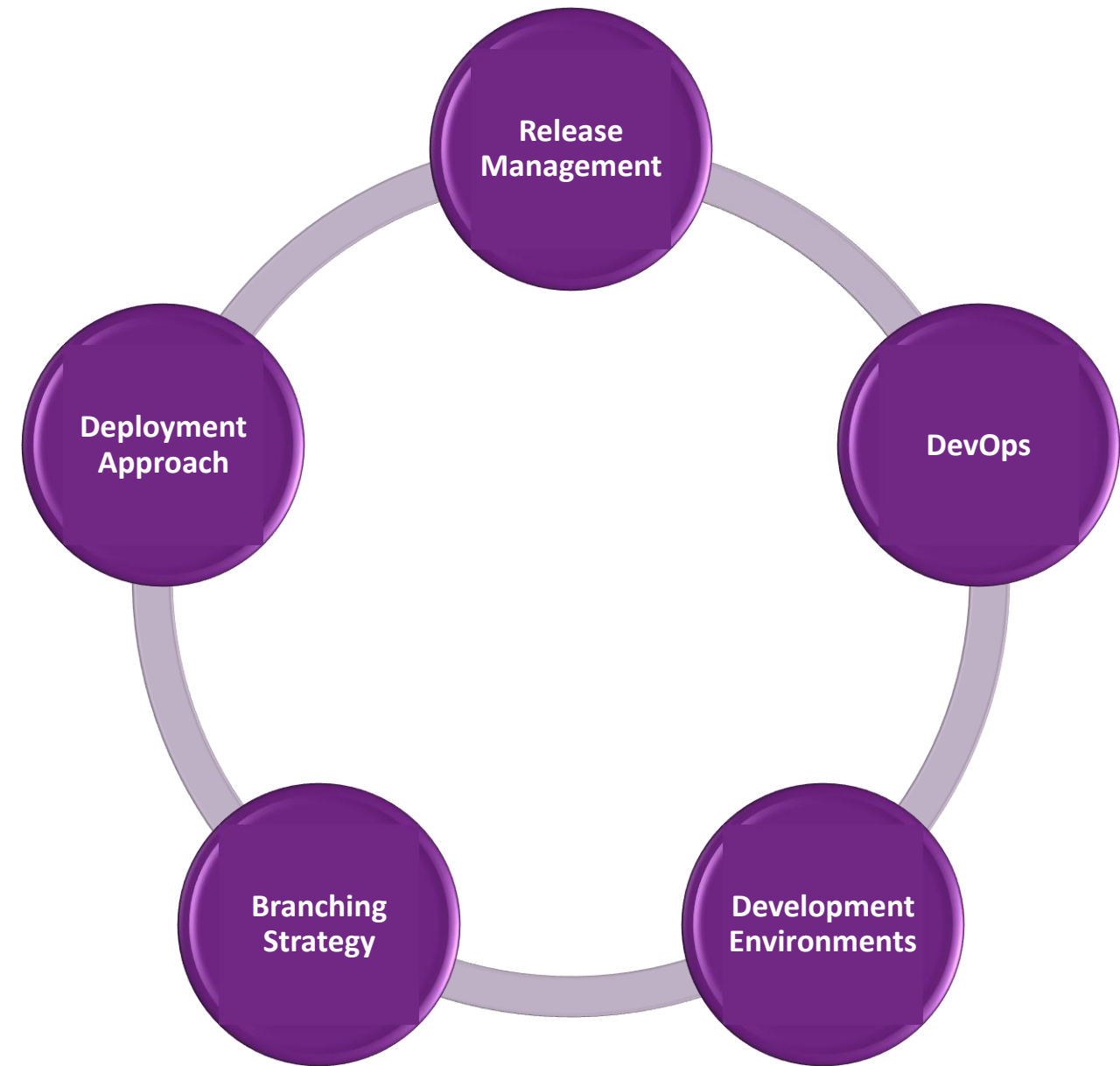
Causes

- Poor Collaboration
- Delays and Inconsistent in Deployments
- Disruptions to Production (Live)
- Poor Data Quality

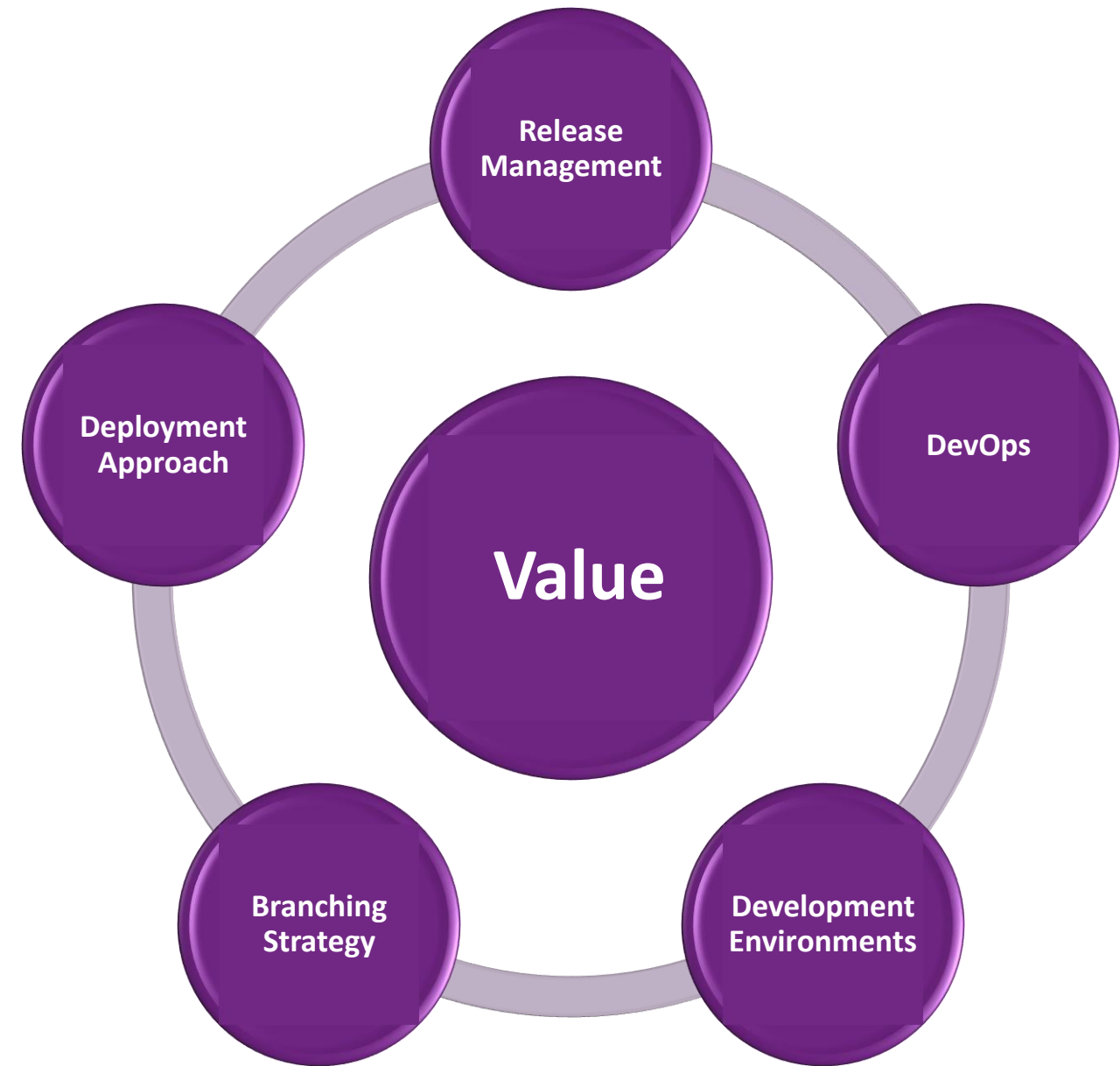


Stakeholders

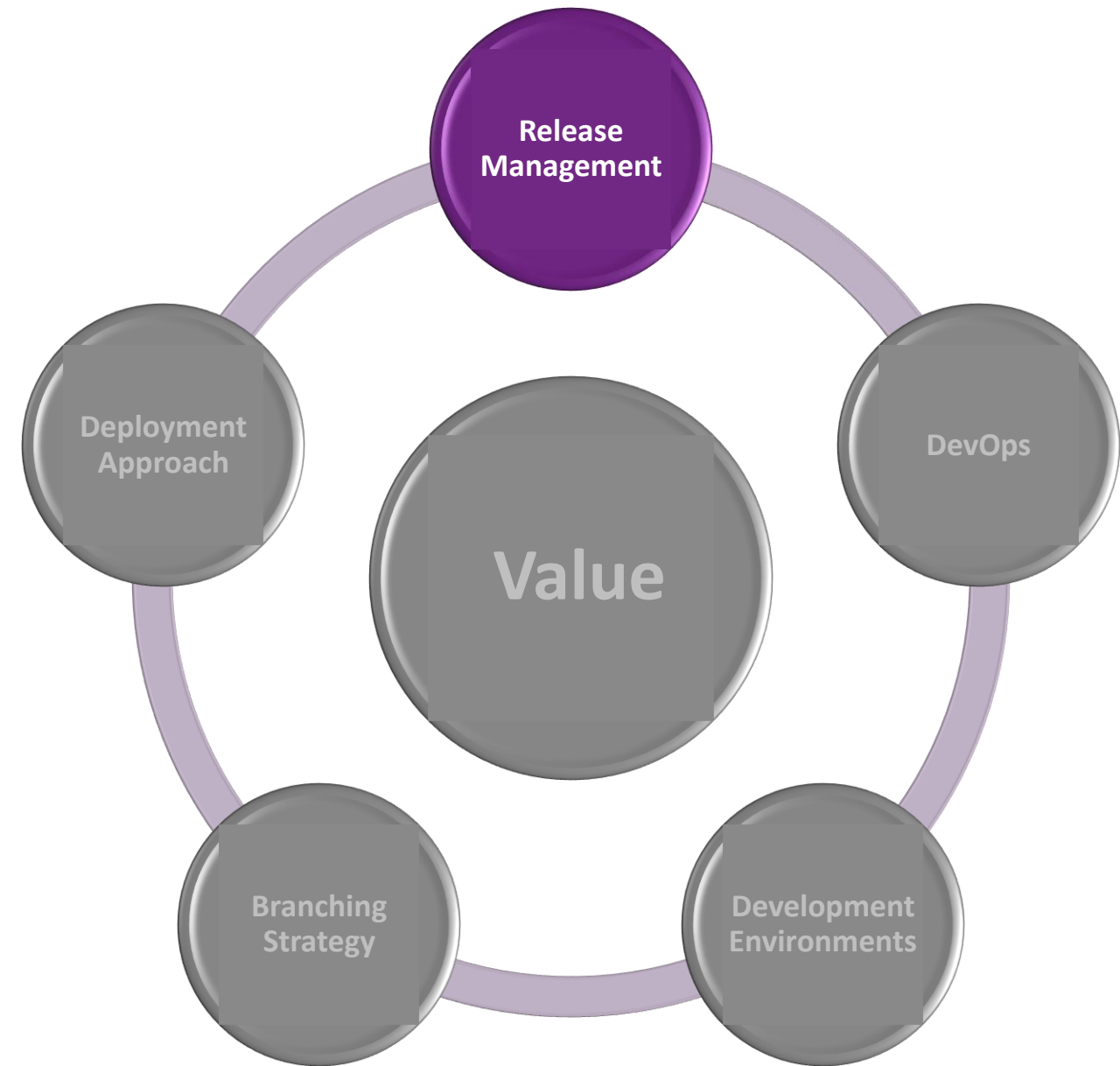




Session Scope

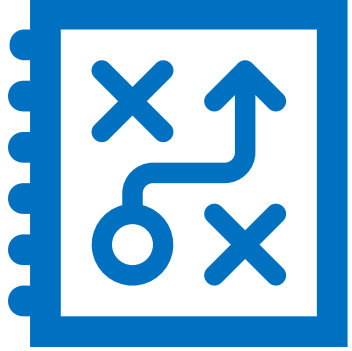


Session Scope



Release Management

What is Release Management



Process of Managing Releases



Lifecycle



Environment



Delivery



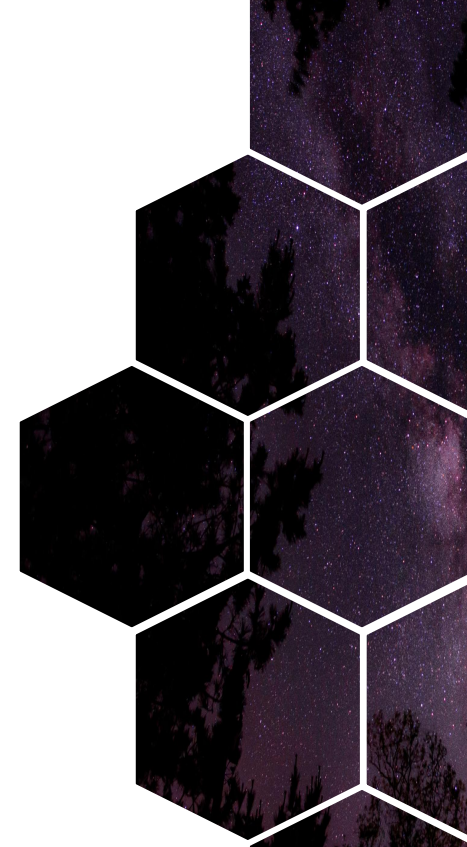
Securely



Efficiently



Users



Release Management - Process



Release Strategy

- Strategic
- Principles and Guidelines
- When, What?
- Impact on Users
- Scope of Change
- Release Lifecycle



Release Plan



Release Management - Process



Release Strategy

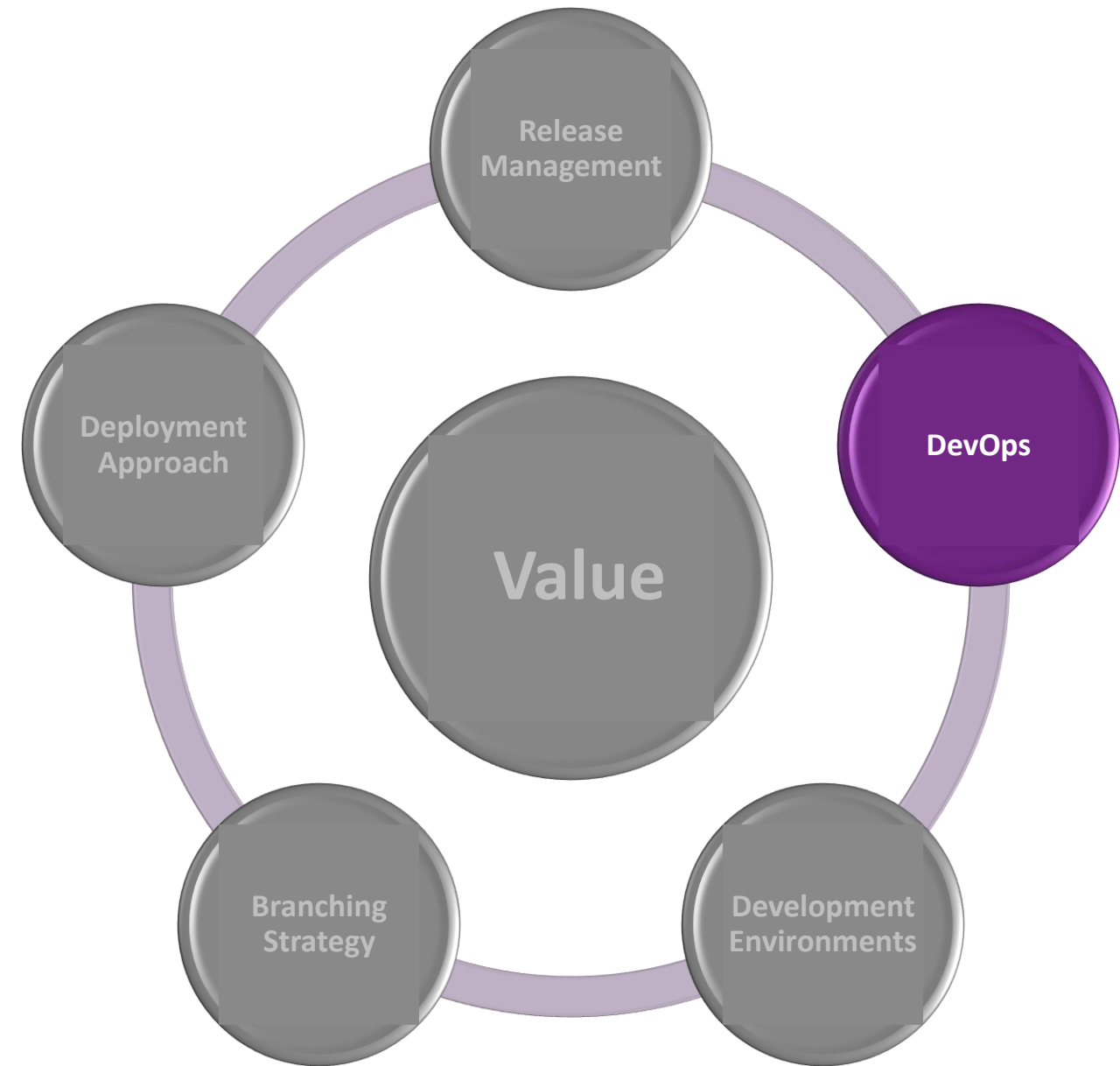
- Strategic
- Principles and Guidelines
- When, What?
- Impact on Users
- Scope of Change
- Release Lifecycle



Release Plan

- Tactical
- Step-by-step Roadmap
- How and Who?
- Impact on Environments
- Release Policies & Approvers
- Consistency



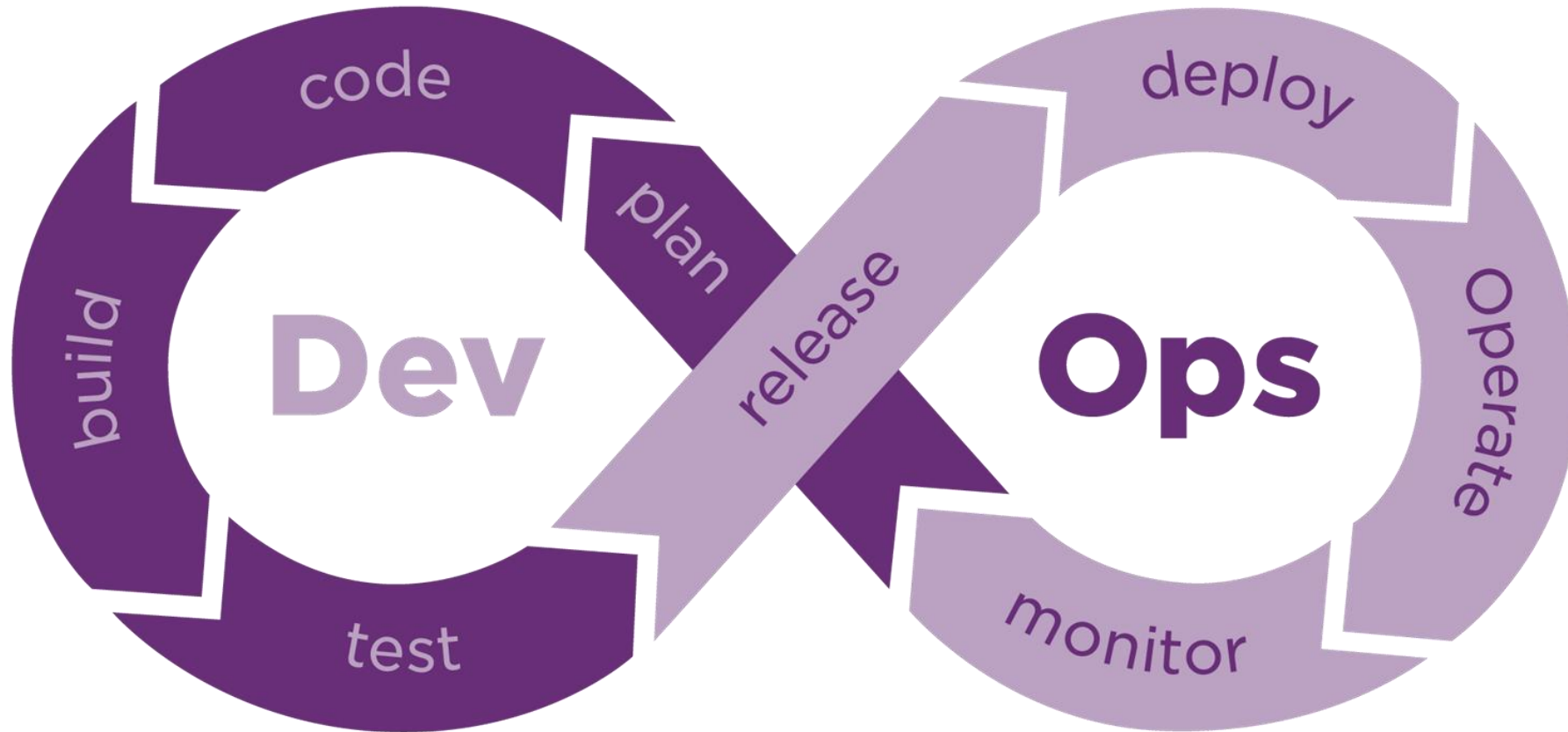


DevOps

What is DevOps



“DevOps is the union of **people**, **process**, and **products** to enable **continuous delivery** of value to our end users.”



DevOps - Delivery



Version Control



Continuous Delivery



Continuous Integration (CI)

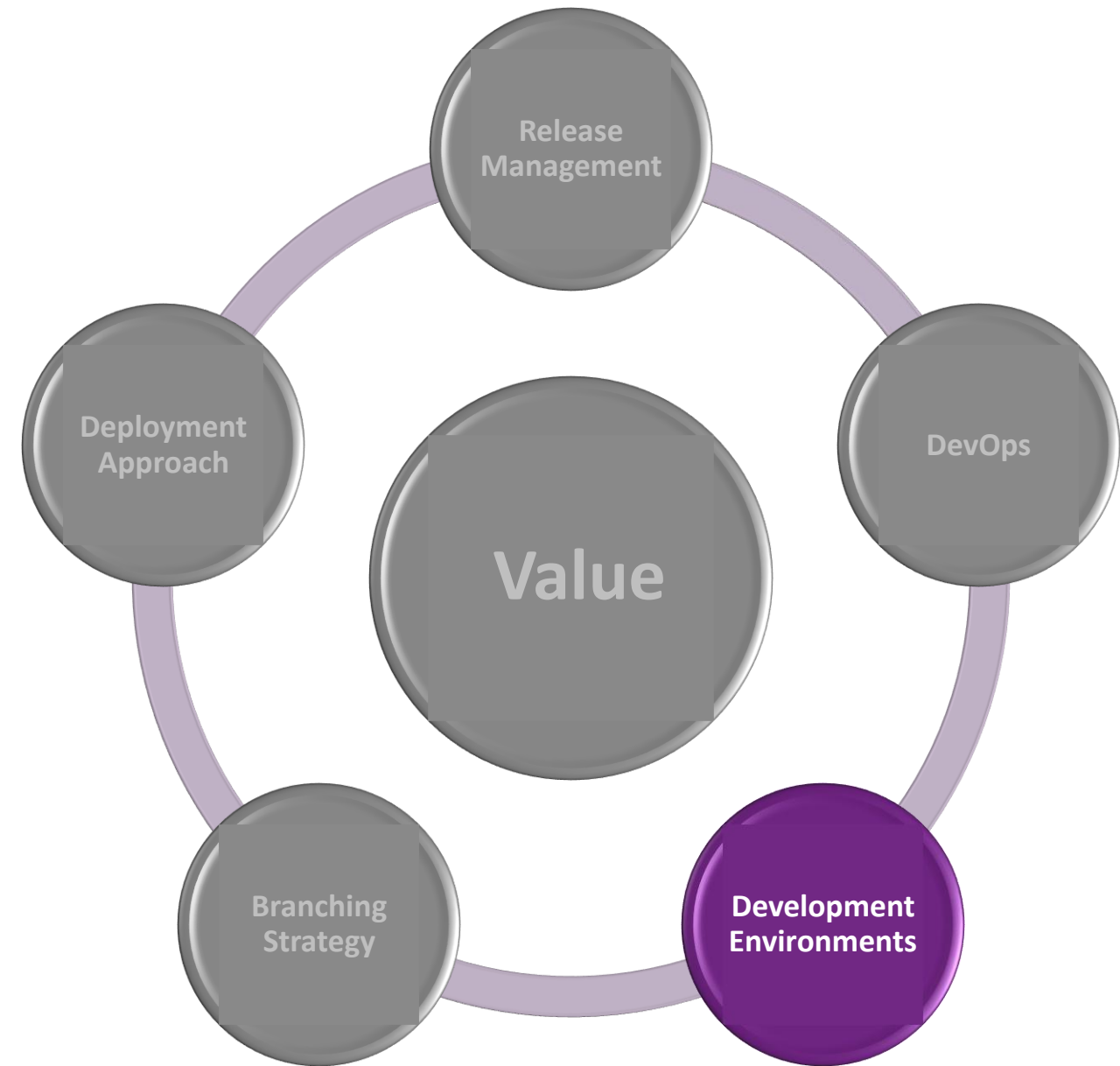
- Improve development
- Build and test solutions
- Detecting bugs
- Rollback



Continuous Deployment (CD)

- Deploy identical solutions
- Code committed to production





Development Environments

Development Environments

Development

Test

Staging / Pre-Prod

Prod



Environments

Development



- Develop, Updates & Commit Code
- No Sensitive & Limited data
- Users: Developer
- Unit/ Integration Testing

Test



- Identity & Fix Bugs
- No Sensitive & Limited data
- Users: Testers & Selected End-Users
- Functional/ Quality Assurance (QA) Testing

Staging / Pre-Prod



- Exact Replica of Production
- Limited Production Data
- Users: Selected End-Users & Release Management
- UAT/ Integration/ Performance Testing
- Final Quality Assurance (QA)

Prod



- Code/ Functionality Published to Live
- Full Production Data
- Users: All End Users
- No Testing



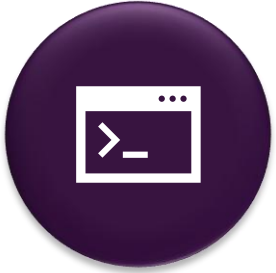
Development Environments

Development

Test

Staging / Pre-Prod

Prod



Identify Bugs & Issues



Reduces Delays



Reduce Risk

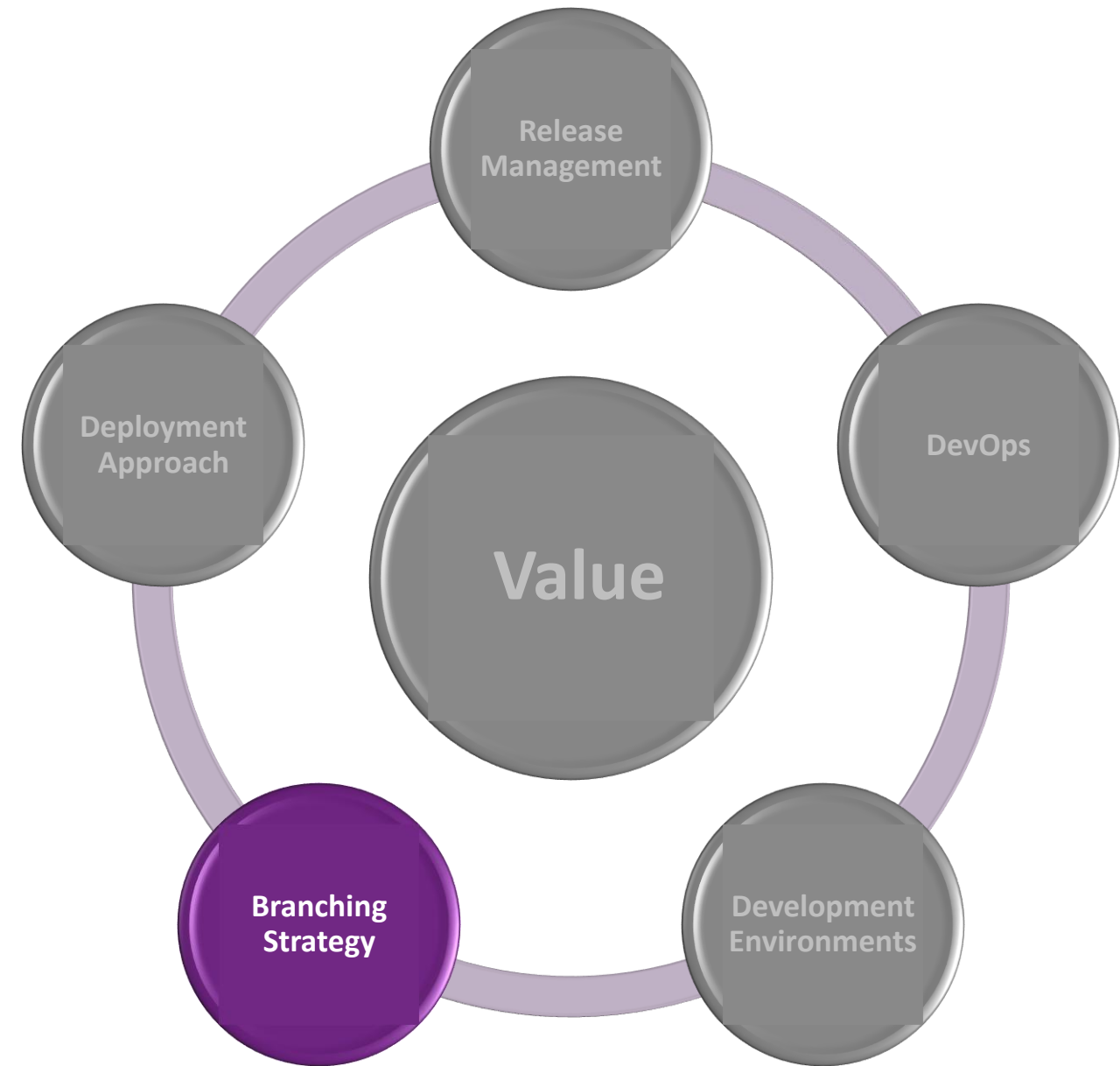


Improve Quality



Improve Collaboration





Branching Strategy

Branching Strategy



Manage Source Code Changes



Track Code Changes



Enables Collaborative



Enables Simultaneously Development



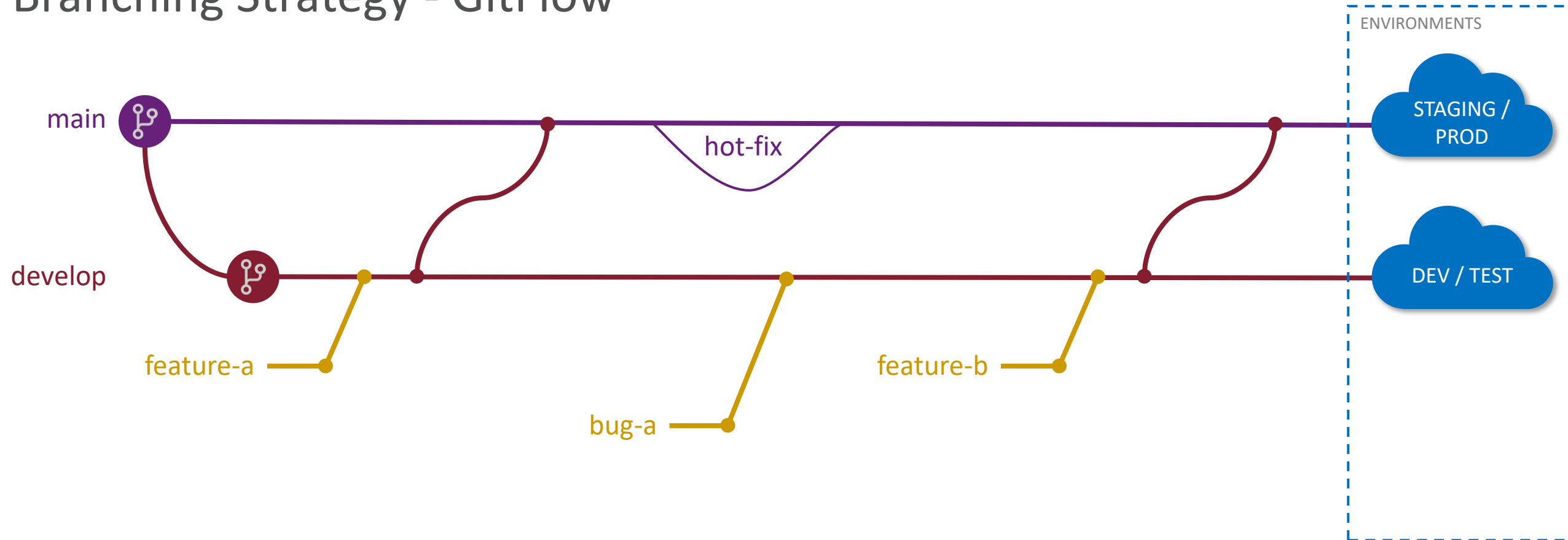
Reduces Developer Risk



Branches can become
Messy and
Unmanageable!!



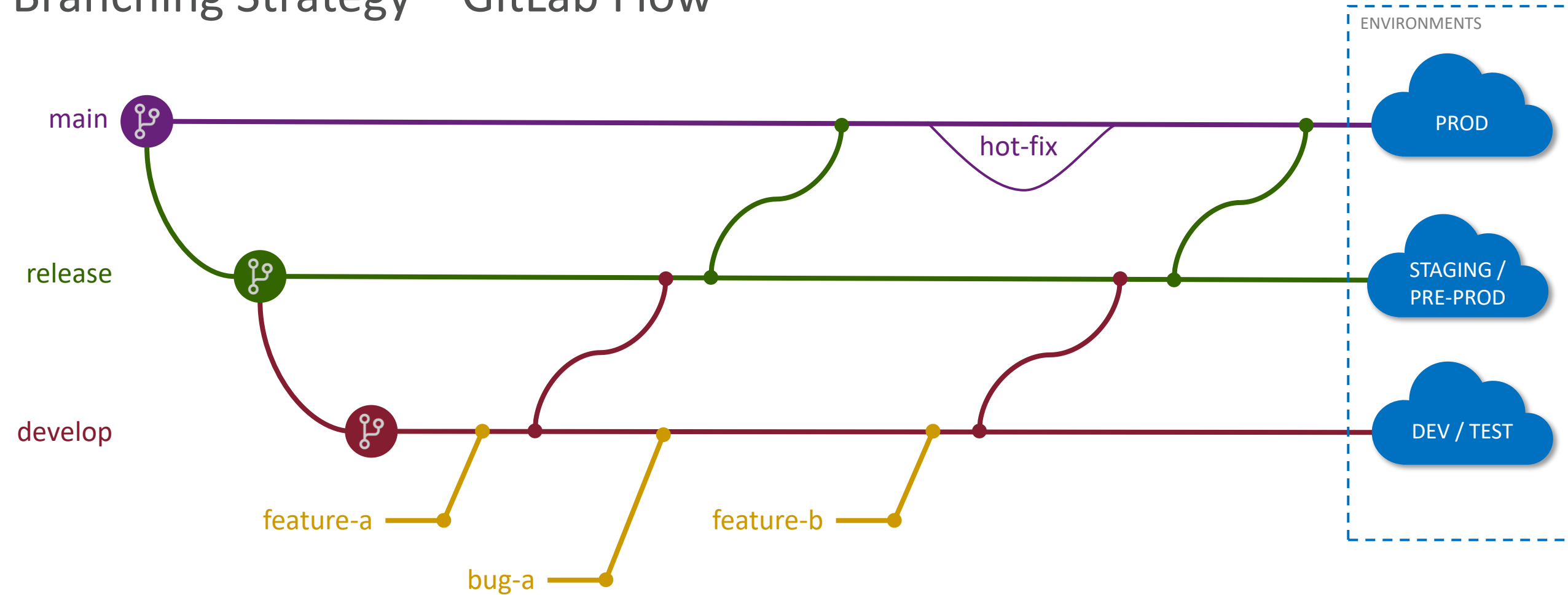
Branching Strategy - GitFlow



- Simple implement and manage
- Multi-branch with specific purposes

- Git history hard to read

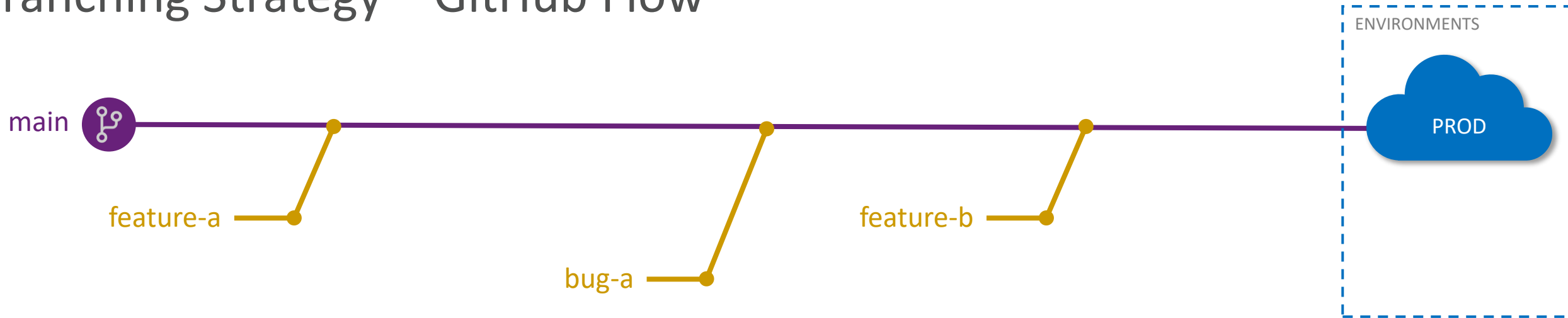
Branching Strategy – GitLab Flow



- Isolation between environments
- Multi-branch with specific purposes
- Git history clean and easy to read

- Complex to implement and manage
- Branches complicated and messy

Branching Strategy – GitHub Flow



- Simpler approach and workflow
- Git history clean and easy to read
- Good for maintain a single production version

- Not ideal for maintaining multiple versions of the code
- Unstable production code

Choosing a Branching Strategy



Project Requirements



Development Method



Scale



Users



CI/CD Tool



Choosing a Branching Strategy



Faster Release Cycles



Optimizes Productivity



Supports Version Control

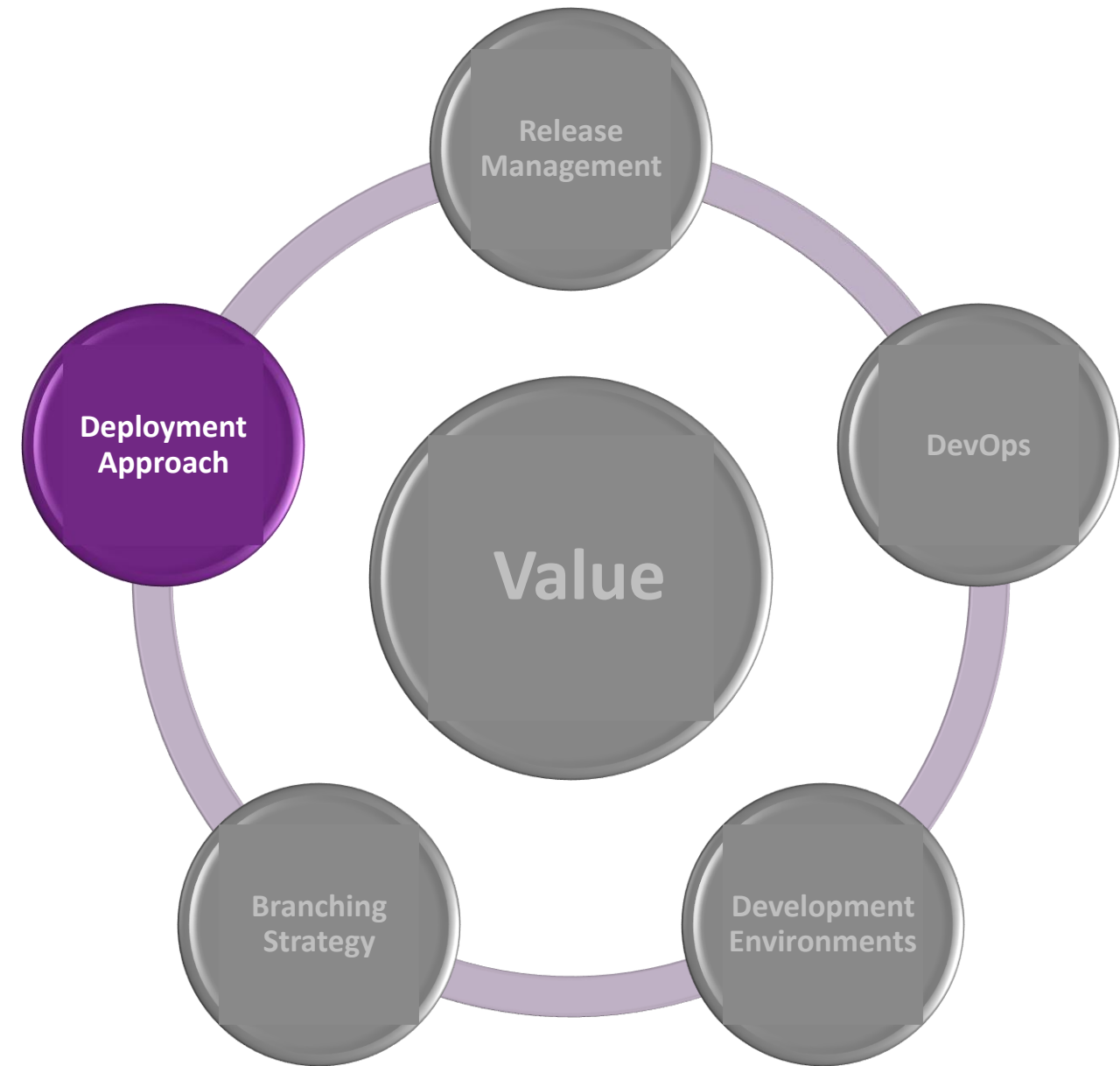


Enables Collaboration



Integrates DevOps Tools





Deployment Approach

Deployment Approach



Series of Deployment Tasks



Without Manual Intervention



Without Explicit Permissions



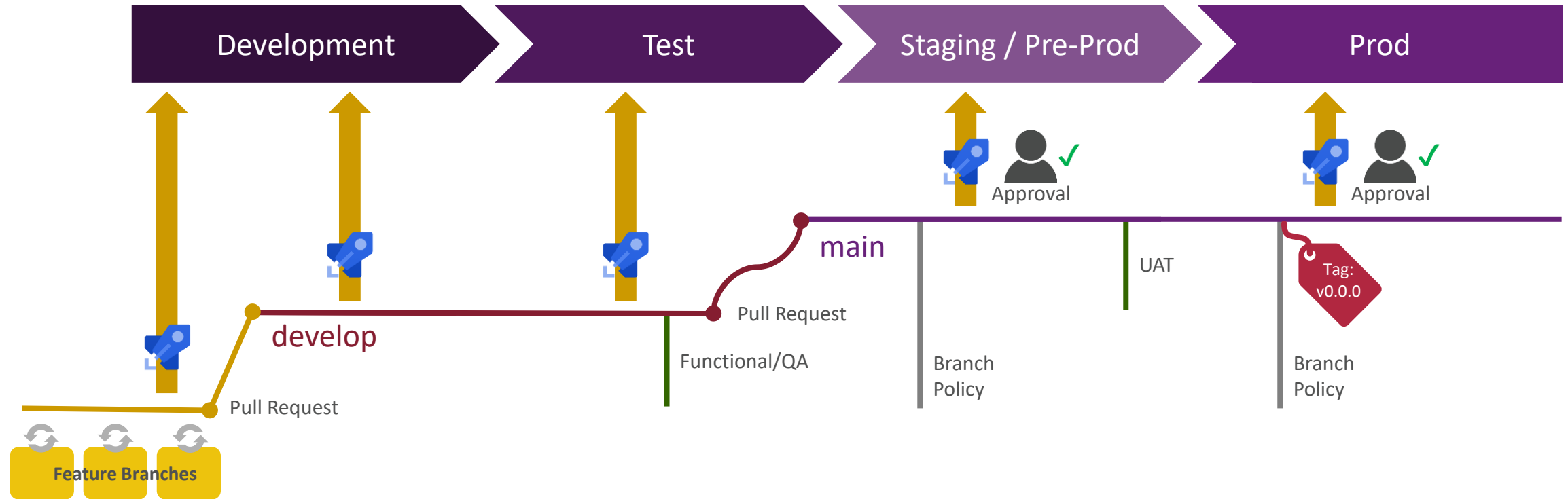
Using CI/CD tool

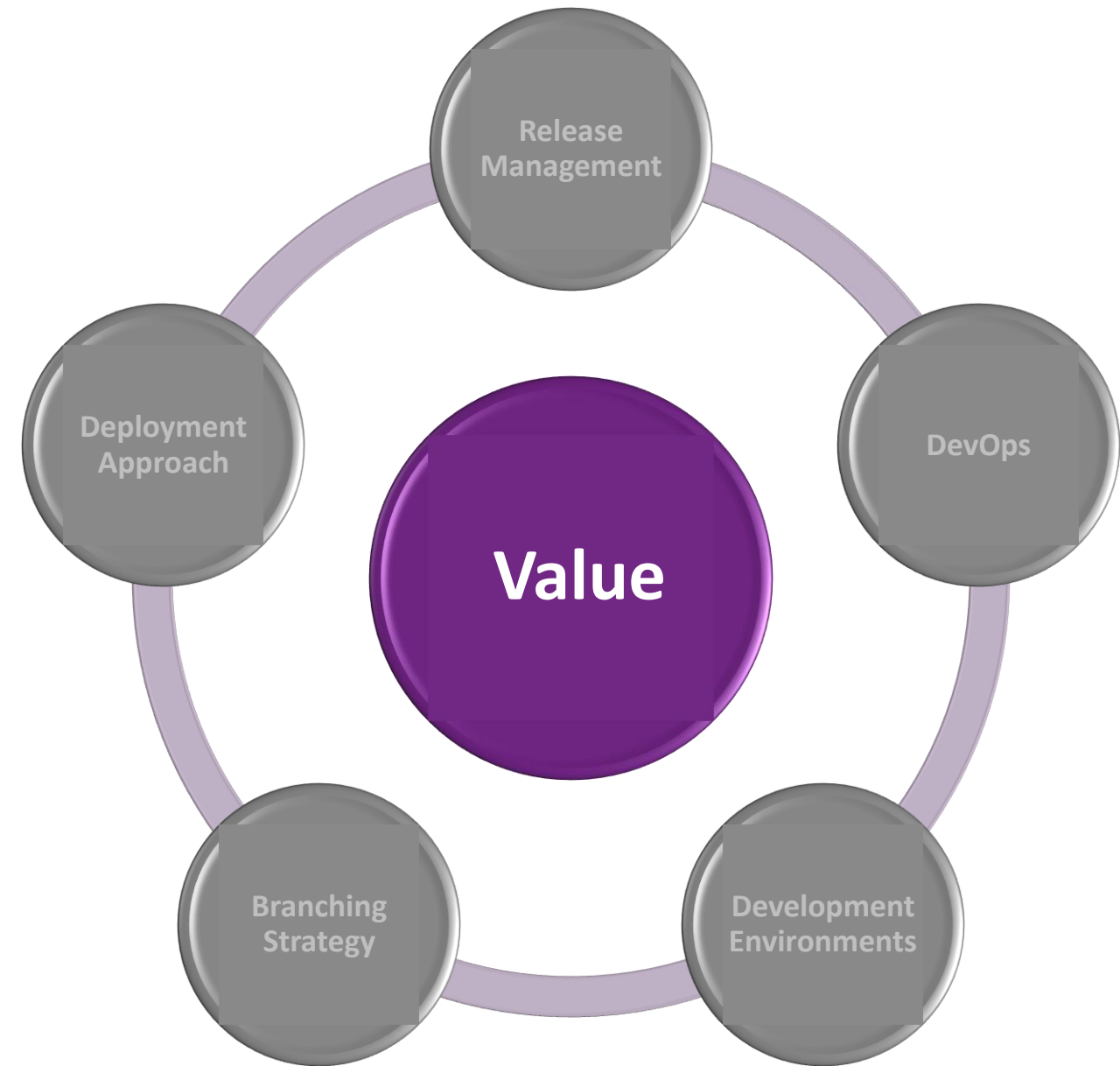


Deployment Tools



Deployment Release Process





Value

The Value

Development Team



- Better Collaboration
- Fewer Single Points of Failures
- More Successful Releases
- Improves Efficiency
- Less Manual Activities
- Identify Bugs Early
- More Feedback

End Users



- Quicker Delivery
- Better Feedback Loop
- Less Downtime, Bugs and Errors
- More Confident in Releases

Business



- More Value Seen in Development Investments
- More Quality and Efficiency Outputs
- Enabling Data Culture
- Accelerated Digital Transformation





Wrap Up

Wrap Up



Breakdown Silos



Fostering Continuous Improvement



Improved Reliability and Stability



Increase Efficiency and Productivity



Reduced Costs



Thank You



Feedback Please!

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